

MSDM5004

Numerical Methods and Modeling in Science

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Lecture 11

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2. Hyperbolic PDEs

Examples

Linear advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Burgers equation

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{2} u^2 \right) = 0$$

Conservation law

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Wave equation

$$\frac{\partial^2 u}{\partial t^2} - \alpha^2 \frac{\partial^2 u}{\partial x^2} = 0$$

2.1. Characteristics and solutions

We focus on the linear advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0 \quad -\infty < x < \infty, \quad t > 0 \quad (1)$$

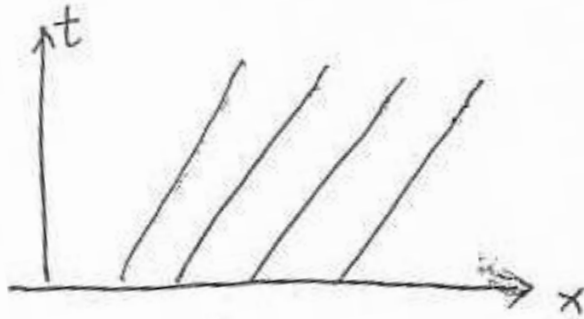
where a is a constant.

The characteristics of Eq. (1) are curves in the (x, t) plane satisfying

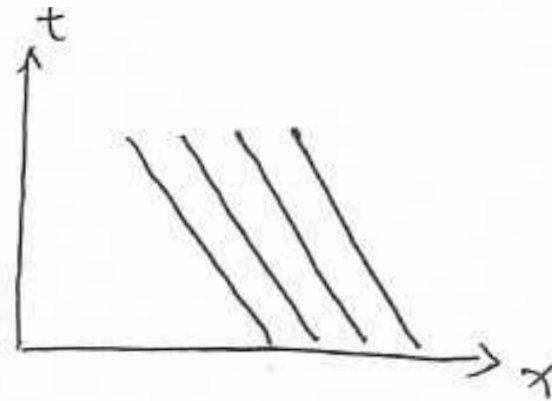
$$\frac{dx(t)}{dt} = a$$

If a is a constant, then the characteristics are straight parallel lines:

$$x - at = \text{constant}.$$



$a > 0$



$a < 0$

Along a characteristic, the solution of Eq. (1) is a constant.

In fact, if $\frac{dx(t)}{dt} = a$, then

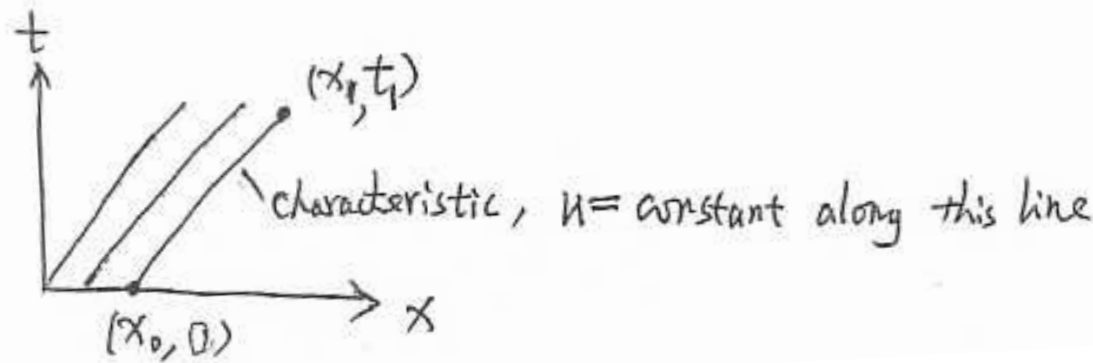
$$\frac{du(x(t), t)}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dt} = \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Thus $u(x(t), t)$ is constant along a characteristic $\frac{dx(t)}{dt} = a$.

The solution of Eq.(1) with initial condition

$$u(x, 0) = u_0(x), \quad -\infty < x < \infty \quad (2)$$

can be obtained using this property.



$$u(x_1, t_1) = u(x_0, 0) = u_0(x_0)$$

Since $x_0 - a \cdot 0 = x_1 - at_1$, we have $x_0 = x_1 - at_1$.

Therefore $u(x_1, t_1) = u_0(x_1 - at_1)$.

That is, the solution of Eq. (1) with initial condition (2) is 5

$$u(x, t) = u_0(x - at).$$

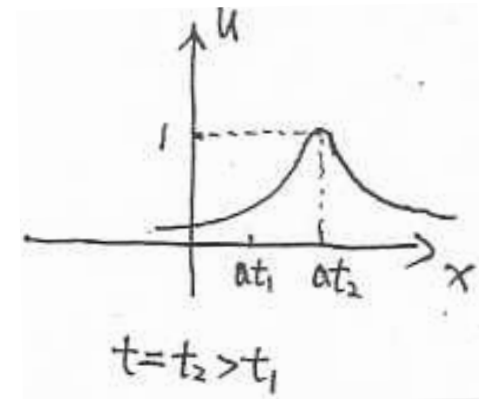
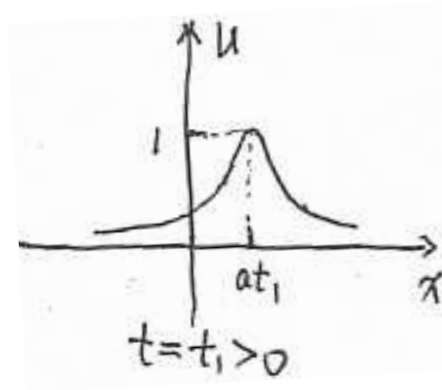
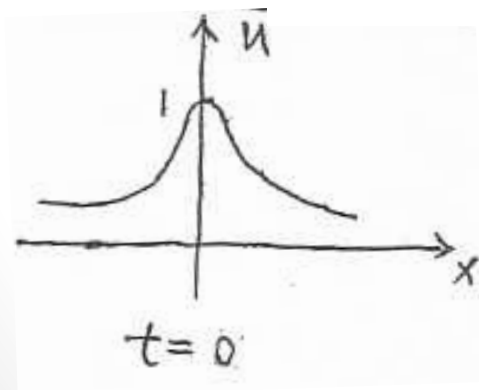
An example

$$u_0(x) = \frac{1}{x^2 + 1}$$

The solution of Eq (1) with this initial condition is

$$u(x, t) = u_0(x - at) = \frac{1}{(x - at)^2 + 1}$$

Solution u at different time (when $a > 0$):



The initial profile moves to the left (if $a > 0$) with the speed a .

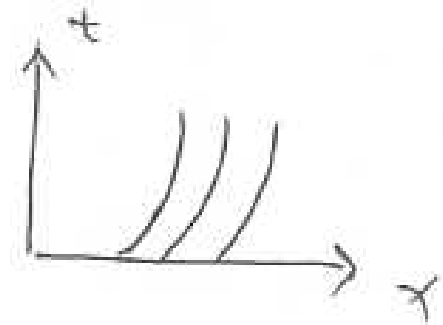
For a general linear PDE

$$\frac{\partial u}{\partial t} + a(x, t) \frac{\partial u}{\partial x} = 0$$

A characteristic $x(t)$ satisfies

$$\frac{dx(t)}{dt} = a(x, t).$$

It is not necessarily a straight line.



For the nonlinear conservation law

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0,$$

it can be written as

$$\frac{\partial u}{\partial t} + f'(u) \frac{\partial u}{\partial x} = 0.$$

A characteristic $x(t)$ satisfies

$$\frac{dx(t)}{dt} = f'(u(x, t)).$$

Characteristics may cross.



2.2. Domain of dependence of the exact solution

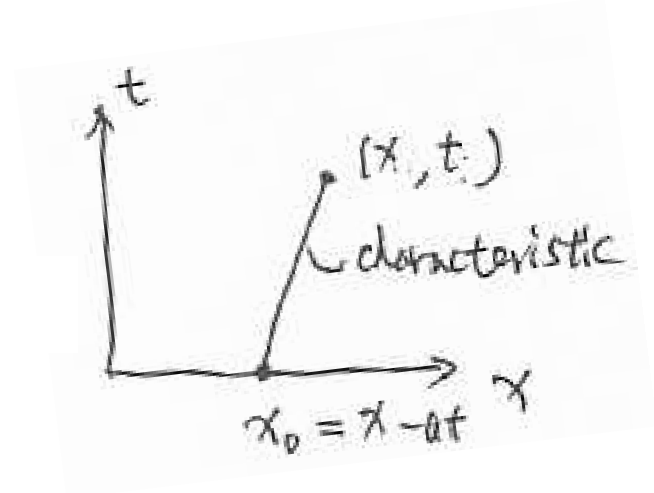
We focus on the linear advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

where a is a constant.

Solution:

$$u(x, t) = u_0(x_0) = u_0(x - at)$$



The solution depends on the initial value u_0 at $x - at$.

Information propagation speed $= a$, which is finite.

Domain of dependence

The set of all points in the space where the initial data at $t=0$ may have some effect on the solution $u(x,t)$.

For

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

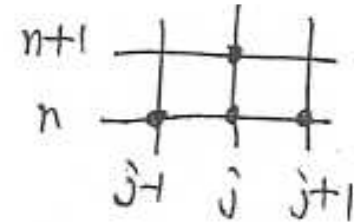
the domain of dependence of the solution $u(x,t)$ is: $\{x-at\}$.

2.3. Some numerical schemes

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Stencil

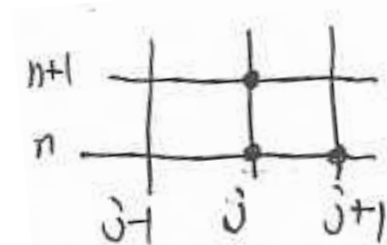
$$(1) \quad \frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x} = 0$$



$$(2) \quad \frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_j^n - U_{j-1}^n}{\Delta x} = 0$$



$$(3) \quad \frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_{j+1}^n - U_j^n}{\Delta x} = 0$$



Let $\nu = \frac{\Delta t}{\Delta x}$.

Note: It is different from $\mu = \frac{\Delta t}{(\Delta x)^2}$ for the diffusion equation.

These schemes can be written as

$$(1) \quad U_j^{n+1} = U_j^n - \frac{a\nu}{2} (U_{j+1}^n - U_{j-1}^n)$$

$$(2) \quad U_j^{n+1} = U_j^n - a\nu (U_j^n - U_{j-1}^n)$$

$$(3) \quad U_j^{n+1} = U_j^n - a\nu (U_{j+1}^n - U_j^n)$$

2.4. Truncation error

$$\begin{aligned} (1) \quad T(x_j, t_n) &= \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} + a \frac{u(x_{j+1}, t_n) - u(x_{j-1}, t_n)}{2\Delta x} \\ &= \frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t + \frac{1}{6} a \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + O((\Delta t)^2) + O((\Delta x)^3). \end{aligned}$$

First order in t , second order in x .

$$\begin{aligned}
 (2) \quad T(x_j, t_n) &= \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} + a \frac{u(x_j, t_n) - u(x_{j-1}, t_n)}{\Delta x} \\
 &= \frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t - \frac{1}{2} a \frac{\partial^2 u}{\partial x^2} \Delta x + O((\Delta t)^2) + O((\Delta x)^2).
 \end{aligned}$$

First order in both t and x .

$$\begin{aligned}
 (3) \quad T(x_j, t_n) &= \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} + a \frac{u(x_{j+1}, t_n) - u(x_j, t_n)}{\Delta x} \\
 &= \frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t + \frac{1}{2} a \frac{\partial^2 u}{\partial x^2} \Delta x + O((\Delta t)^2) + O((\Delta x)^2).
 \end{aligned}$$

First order in both t and x .

$$k = \bar{k} \frac{2\pi}{L}, \bar{k} \text{ is an integer}$$

2.5. Stability

Fourier analysis

$$U_j^n = [\lambda(k)]^n e^{ikx_j}$$

$$(1) \quad \lambda(k) = 1 + ia\nu \sin k\Delta x$$

$$|\lambda(k)|^2 = 1 + a^2\nu^2 \sin^2 k\Delta x$$

$$\text{When } k\Delta x = \frac{\pi}{2}, \quad |\lambda(k)|^2 = 1 + a^2\nu^2 > 1.$$

Thus $|\lambda(k)|^n$ is unbounded for large n .

This scheme (central difference scheme) is always unstable.

$$(2) \quad \lambda(k) = 1 - a\nu (1 - e^{-ik\Delta x})$$

$$|\lambda(k)|^2 = 1 - 4a\nu(1 - a\nu) \sin^2 \frac{k\Delta x}{2}$$

This scheme is stable if and only if $|\lambda(k)| \leq 1$, which is equivalent to

$$a\nu(1 - a\nu) \geq 0.$$

$a > 0$, stability condition $a\nu \leq 1$.

$a < 0$, always unstable.

$$(3) \quad \lambda(k) = 1 - a\nu (e^{ik\Delta x} - 1)$$

$$|\lambda(k)|^2 = 1 + 4a\nu(1 + a\nu) \sin^2 \frac{k\Delta x}{2}$$

$a > 0$, always unstable.

$a < 0$, stability condition $|a|\nu \leq 1$.

Therefore,

- When $a > 0$, use scheme (2);
- When $a < 0$, use scheme (3).

This scheme is called upwind scheme.

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

The upwind scheme is

$$\begin{cases} \frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_j^n - U_{j-1}^n}{\Delta x} = 0, & \text{if } a > 0 \\ \frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_{j+1}^n - U_j^n}{\Delta x} = 0, & \text{if } a < 0 \end{cases}$$

or

$$\begin{cases} U_j^{n+1} = U_j^n - a\nu (U_j^n - U_{j-1}^n), & \text{if } a > 0 \\ U_j^{n+1} = U_j^n - a\nu (U_{j+1}^n - U_j^n), & \text{if } a < 0 \end{cases}$$

First order in both t and x .

Stability condition $|a|\nu \leq 1$.